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# LOW-RANK ADAPTIVE FRICTION FOR MEMORY-EFFICIENT LLM PRETRAINING

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PREPRINT

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## ABSTRACT

iKFAD is a recently proposed optimizer that closes the Adam–SGD performance gap observed in transformer model pretraining by replacing adaptive learning rates with adaptive friction in the momentum dynamics. Its limitation is that the full friction tensor  $\xi \in \mathbb{R}^{m \times n}$  carries the same  $\mathcal{O}(nm)$  memory overhead per layer as Adam’s second-moment buffer. Here we exploit the rank-1 compression idea that Adafactor applies to Adam’s second moment, replacing iKFAD’s friction tensor  $\xi$  with a rank-1 outer-product factorization built from row and column momentum statistics, resulting in Rank-1 iKFAD (R-iKFAD). This reduces friction memory footprint from  $\mathcal{O}(nm)$  to  $\mathcal{O}(n + m)$  per layer which approximately halves iKFAD’s total optimizer state. Despite this massive decrease in signal, we show that R-iKFAD maintains parity in performance with iKFAD, and consequently Adam on transformer pretraining tasks. Experiments on GPT2-Nano, TinyViT, DistilBERT, and GPT2-S confirm that R-iKFAD matches or exceeds iKFAD while nearly halving the memory footprint and being comparably robust to hyperparameters. We analyze the continuous-time dynamics in two damping regimes. For  $\gamma > 0$  (linear damping) we prove exponential convergence under strong convexity. For  $\gamma = 0$ , the preferred option in our experiments, the friction is generated entirely based on past momentum and it switches off as the momentum vanishes, so geometric convergence cannot be shown. We nonetheless prove unconditional convergence to the minimizer together with explicit algebraic rates,  $\mathcal{O}(t^{-1})$  and  $\mathcal{O}(t^{-1/2})$  according to the regularization scale. To our knowledge this is the first convergence rate for a rank-1 factored optimizer in continuous time, and the first such result that does not require positive damping.

## 1 Introduction

Adam [10] is the workhorse optimizer in modern deep learning, combining gradient momentum with coordinate-wise second-moment scaling to accelerate convergence on complex loss landscapes. For a model with  $N$  scalar parameters, Adam evolves two auxiliary states (the first and second moment estimates) and therefore requires  $2N$  auxiliary scalars on top of the  $N$  parameters themselves, for a total memory cost of around  $3N$ . At scale, this  $2N$  overhead presents a significant hardware bottleneck during large-model training. To mitigate this, Adafactor [17] introduces a rank-1 factorization of the second-moment matrix. For a matrix weight  $W \in \mathbb{R}^{m \times n}$ , Adafactor replaces Adam’s full second-moment matrix  $V \in \mathbb{R}^{m \times n}$  with a factorized approximation  $\hat{V}_{ij} = R_i C_j / (\mathbf{1}_m^\top R)$ , where  $R \in \mathbb{R}^m$  and  $C \in \mathbb{R}^n$  track row-wise and column-wise squared gradients, respectively. By eliminating the first-moment buffer, Adafactor reduces per-layer auxiliary state complexity from  $\mathcal{O}(mn)$  to  $\mathcal{O}(m + n)$  (achieving memory savings of over 99% for large weight matrices) while maintaining competitive transformer pretraining performance.

An insightful paradigm for analyzing optimizer dynamics is based on continuous-time ordinary differential equations (ODEs) [3, 5], where updates emerge as discrete time-step approximations. For parameter state  $x$ , momentum  $p$ , and second-moment variable  $\zeta$ , continuous-time Adam is given in Eqs. (1)–(3). While Adam incorporates parameter-wise

adaptation into the *position* equation ( $\dot{x}$ ) via  $1/\sqrt{\zeta}$ , Karani et al. [7] demonstrated that coordinate-wise adaptation can alternatively be integrated into the *momentum* equation ( $\dot{p}$ ). This is achieved by replacing the constant damping factor  $\gamma$  with a dynamic, coordinate-wise friction state  $\xi$ , resulting in the Individual Kinetic Friction-Adaptive Descent (iKFAD) dynamics (4)-(6)

$$\dot{x} = \frac{p}{\sqrt{\zeta} + \epsilon}, \quad (1) \quad \dot{x} = p, \quad (4)$$

$$\dot{p} = -\nabla f(x) - \gamma p, \quad (2) \quad \dot{p} = -\nabla f(x) - (\gamma + \xi) \odot p, \quad (5)$$

$$\dot{\zeta} = [\nabla f(x)]^2 - \alpha \zeta. \quad (3) \quad \dot{\xi} = \frac{[p]^2}{\mu} - \alpha \xi. \quad (6)$$

These dynamics require storing a full adaptive friction tensor  $\xi$  alongside momentum  $p$ . They incur an auxiliary footprint of  $2N$  scalars, identical to Adam’s auxiliary storage. Originally introduced for scalar/global friction [6], iKFAD extends friction-based adaptation to individual parameters, matching Adam’s performance on standard vision and NLP benchmarks.

## 2 Methodology: Rank-1 iKFAD

Adafactor compresses Adam’s second moment; Rank-1 iKFAD (R-iKFAD) applies the same compression to iKFAD’s friction. Writing both as continuous-time systems for a matrix weight  $X \in \mathbb{R}^{m \times n}$  makes the correspondence apparent, in the same sense in which the systems above approximate Adam and iKFAD [3, 5]. Most Adafactor implementations omit the first moment, but we enable it throughout for parity, so we retain momentum  $P$  and refer to the result as **Adafactor-m** (left). R-iKFAD (right) carries the same momentum and factorizes the friction instead:

$$\dot{X} = \frac{P}{\sqrt{\hat{\zeta}} + \epsilon}, \quad (7) \quad \dot{X} = P, \quad (11)$$

$$\dot{P} = -\nabla f(X) - \gamma P, \quad (8) \quad \dot{P} = -\nabla f(X) - \tilde{\xi} \odot P - \gamma P, \quad (12)$$

$$\dot{R} = [\nabla f(X)]^2 \mathbf{1}_n - \alpha R, \quad (9) \quad \dot{R} = \frac{1}{\mu} [P]^2 \mathbf{1}_n - \alpha R, \quad (13)$$

$$\dot{C} = \mathbf{1}_m^\top [\nabla f(X)]^2 - \alpha C. \quad (10) \quad \dot{C} = \frac{1}{\mu} \mathbf{1}_m^\top [P]^2 - \alpha C. \quad (14)$$

Here  $\odot$  is the element-wise product and  $[P]_{ij}^2 = P_{ij}^2$ ; both systems carry the same linear damping  $\gamma$ . Each builds a rank-1 object from its factors,

$$\hat{\zeta} = \frac{RC^\top}{\mathbf{1}_m^\top R}, \quad \tilde{\xi}_{ij} = \frac{R_i C_j}{\mathbf{1}_m^\top R + \epsilon_{\text{stab}}}, \quad (15)$$

with  $\epsilon, \epsilon_{\text{stab}} > 0$  small regularizers, the latter ensuring numerical stability near  $R = 0$ . Adafactor-m is thus Adam’s system with the full second moment  $\zeta$  replaced by the rank-1 estimate  $\hat{\zeta}$ , and R-iKFAD is iKFAD’s system with the full friction tensor  $\xi$  replaced by the rank-1 estimate  $\tilde{\xi}$ . The two differ in the same way Adam differs from iKFAD: Adafactor-m’s factors aggregate squared *gradients* and act in the position equation, whereas R-iKFAD’s aggregate squared *momentum* and act as damping in the momentum equation. In both cases the state per layer falls from  $\mathcal{O}(mn)$  to  $\mathcal{O}(m+n)$  while full momentum is retained. We emphasize that R-iKFAD is designed to maintain parity with iKFAD (which matches Adam [7] in performance), rather than outperforming it.

**Splitting Discretization.** To construct a stable discrete-time algorithm, we rely on principles of geometric numerical integration of Hamiltonian systems [12], which have proved a robust foundation for discretization of dynamics-based optimization schemes. Following [7], we decompose the vector field into sub-flows that admit exact analytical integration. The detailed sub-flow decomposition and closed-form update equations are provided in Appendix A.

**Memory Complexity.** For a network parameter matrix at layer  $\ell$  given by  $W_\ell \in \mathbb{R}^{m_\ell \times n_\ell}$ , let  $N = \sum_\ell m_\ell n_\ell$  denote the total number of scalar model parameters. Table 1 breaks down the auxiliary optimizer memory footprint required on top of the model weights. For non-2D parameters (biases, LayerNorm scales) that are not rank-1 factorizable, R-iKFAD falls back to element-wise iKFAD, contributing a negligible additional scalar per parameter. R-iKFAD requires approximately  $N$  auxiliary scalars in total, effectively halving the optimizer auxiliary memory footprint relative to both iKFAD and Adam. Adafactor-m occupies this same  $\approx N$  regime and is therefore our primary baseline; the

difference is which buffer is compressed, the second moment for Adafactor-m versus the friction tensor for R-iKFAD. Retaining momentum is deliberate: Adafactor is known to suffer training instabilities in certain transformer regimes when momentum is discarded entirely [17].

Table 1: Comparison of auxiliary optimizer memory complexity (excluding parameters).

| Optimizer                          | Auxiliary State Variables                      | Total Auxiliary Footprint                 |
|------------------------------------|------------------------------------------------|-------------------------------------------|
| <b>Adam</b>                        | First moment $M$ , Second moment $V$           | $2N$ scalars                              |
| <b>iKFAD</b>                       | Momentum $P$ , Friction tensor $\xi$           | $2N$ scalars                              |
| <b>Adafactor-m, R-iKFAD (ours)</b> | Momentum $P$ , Factors $(R_\ell, C_\ell)_\ell$ | $N + \sum_\ell (m_\ell + n_\ell)$ scalars |

### 3 Theoretical Results

We analyze the continuous-time system (11)–(14) in both damping regimes; statements, proofs and numerical verification are in Appendix E.

For  $\gamma > 0$  a perturbed-energy argument gives exponential convergence under strong convexity (Theorem 2). Every experiment here, however, uses  $\gamma = 0$ . In R-iKFAD,  $\tilde{\xi} \geq 0$  entrywise, and  $\dot{\mathcal{E}} = -\langle P, \tilde{\xi} \odot P \rangle_F \leq 0$ , so at  $\gamma = 0$  the system remains dissipative and still converges to the minimizer. What changes is the character of the dissipation: the friction is no longer imposed externally but manufactured by the trajectory out of its own past momentum. Two consequences follow. The dissipation weakens as  $P \rightarrow 0$ , so no exponential rate can hold, and the decay must be algebraic. Moreover  $\dot{\mathcal{E}}$  vanishes at every turning point of the oscillation, so estimates must be made over sequential time blocks.

We prove two results. Convergence is unconditional: for any coercive  $f$  with a unique critical point, every trajectory reaches the minimizer, for all  $\alpha, \mu, \epsilon_{\text{stab}}$  (Theorem 3). Under uniform strong convexity we also obtain an algebraic rate,

$$\mathcal{E}(t) = \mathcal{O}(t^{-1}) \quad (\epsilon_{\text{stab}} = 0), \quad \mathcal{E}(t) = \mathcal{O}(t^{-1/2}) \quad (\epsilon_{\text{stab}} > 0),$$

with every constant given explicitly (Theorem 8). The two exponents can be viewed as regimes of a single estimate, separated at  $\mathcal{E} \approx \alpha\mu\epsilon_{\text{stab}}$ , so the  $t^{-1}$  law is the operative one at any reachable horizon. We then confirm the theoretical result using numerical experiments, which verify both the exponents and the crossover scale: integrating to  $T = 10^7$ , local log–log slopes over the final decade  $[10^6, 10^7]$  fall within 0.011 of the predicted  $-1$  and within 0.003 of the predicted  $-1/2$ , with the limiting exponents independent of  $m$  and  $n$  (Appendix E.3.6).

To our knowledge this is the first convergence *rate* for a rank-1 factored optimizer in continuous time, and the first analysis of one that does not require positive damping. Appendix F compares it to the vanishing-damping literature and earlier factored-optimizer analyses. The gap being closed also resolves an outstanding issue for iKFAD. iKFAD is suggested to be deployed at  $\gamma = 0$ , yet the existing convergence analysis (see [7]) assumes  $\gamma > 0$ , so the friction-based optimizers have so far lacked a complete convergence theory. The argument is also not tied to the rank-1 structure, and serves as a template for closing the same gap for full-friction iKFAD. At  $m = n = 1$  with  $\epsilon_{\text{stab}} = 0$  the reduction is exact: trace matching gives  $R \equiv C$ , hence  $\tilde{\xi} = R^2/R = R$ , and the system becomes scalar KFAD [6], so the  $t^{-1}$  rate covers that method too.

### 4 Experiments

We set  $\gamma = 0$  for both iKFAD and R-iKFAD throughout (denoted **iKFAD0** and **R-iKFAD0** respectively), following [7]; Section 4.2 confirms  $\gamma > 0$  yields no benefit. We compare both against Adam and Adafactor-m across four benchmarks spanning image classification and language modeling at scales from 0.85M to 125M parameters:

- **GPT2-Nano (0.85M)** [8] on Shakespeare character-level language modeling.
- **TinyViT (5M)** [20] on CIFAR-10 [11] image classification.
- **DistilBERT (80M)** [16] on SST-2 [18] sentiment classification (pretraining from scratch).
- **GPT2-S (125M)** [15] on OpenWebText pretraining (nanoGPT architecture) [8].

To rigorously isolate the effects of the optimization dynamics and ensure a fair comparison, we use a constant learning rate, disable weight decay, and omit additional regularizers across all runs. Furthermore, we enable Adafactor’s optional

first moment ( $\beta_1$  tuned) rather than the conventional momentum-free configuration, that is, we use Adafactor-m, so that it matches R-iKFAD’s auxiliary-memory regime and the comparison is apples-to-apples. All hyperparameters are tuned via Optuna [1] (TPE sampler) with equal sweep budgets of approximately 80 trials per optimizer per model; GPT2-Nano and TinyViT received substantially more trials. See Appendix B for full configurations. Just like iKFAD, the optimal scale of  $\mu$  varies by several orders of magnitude between different tasks due to differing momentum scales. Therefore, we recommend a wide-range initial sweep over  $\mu$  and accept this as a potential limitation of our method.

#### 4.1 Main Results

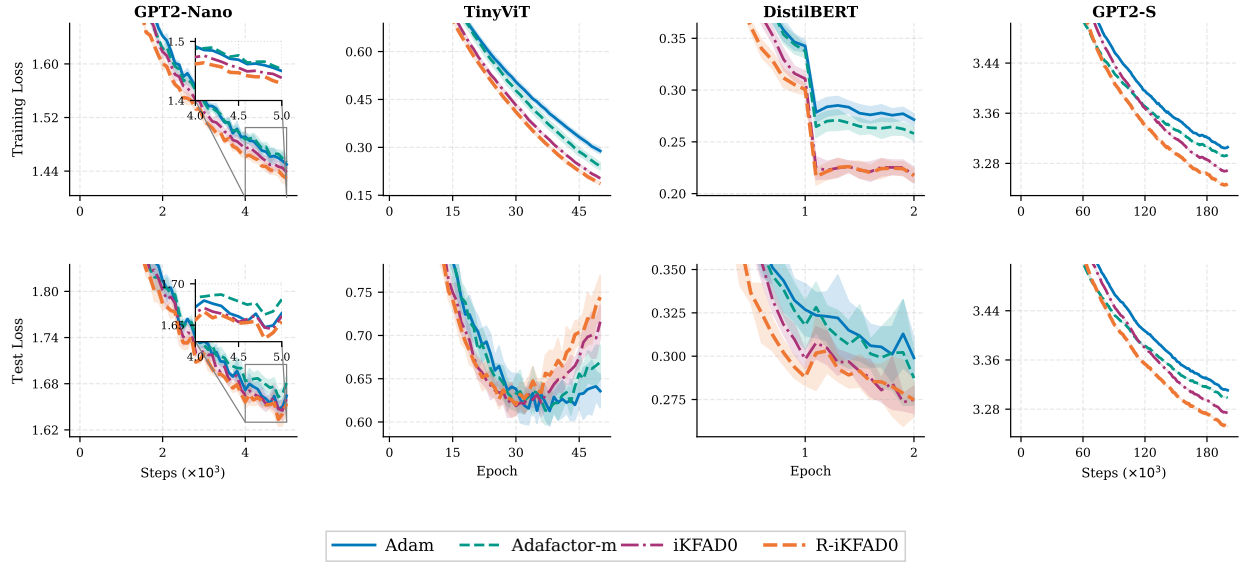


Figure 1: Loss curves for all four benchmarks, averaged over 10 random seeds. R-iKFAD0 matches or improves upon iKFAD0 on all tasks, while using approximately half the optimizer memory. Both methods are competitive with Adam and Adafactor-m, which require comparable or greater memory. Shaded regions denote one standard deviation.

Figures 1 and 2 show training and test loss together with accuracy across all benchmarks. R-iKFAD0 matches iKFAD0 in convergence speed and best achieved performance while using about half the optimizer memory, confirming that the rank-1 friction approximation incurs no performance cost. Crucially, R-iKFAD0 is competitive with Adam throughout, consistent with the finding in [7] that adaptive friction achieves parity with adaptive learning rates on LLM pretraining. We do not claim R-iKFAD surpasses Adam. The goal is to preserve iKFAD’s performance parity with Adam while reducing memory overhead. Table 2 summarizes the lowest test loss achieved during training for all methods. Note that Adafactor-m requires tuning only two hyperparameters, whereas Adam, iKFAD, and R-iKFAD0 all require three. The fixed trial budget hence favored Adafactor-m’s results disproportionately. GPT2-S results use a single seed due to computational constraints; all other benchmarks report mean  $\pm$  standard deviation over 10 seeds. The optimizer auxiliary state is the memory an optimizer adds on top of the model parameters. Table 5 in Appendix D reports the absolute values on all four benchmarks in MB (float32).

#### 4.2 Ablation: $\gamma = 0$ versus $\gamma > 0$

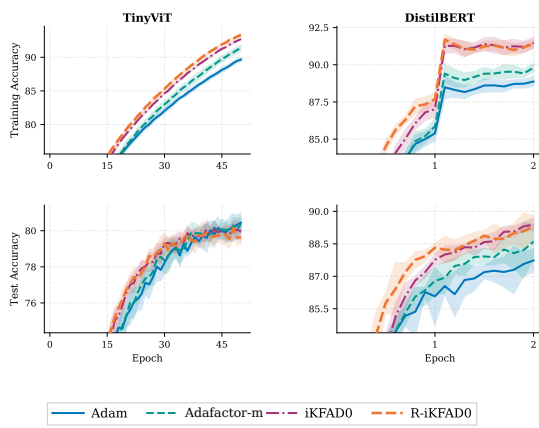
Figure 3 (Appendix C) shows that  $\gamma > 0$  yields no consistent improvement over  $\gamma = 0$  on any of the three benchmarks tested (GPT2-Nano, TinyViT, DistilBERT). This is consistent with the finding in [7] that the adaptive friction mechanism alone provides sufficient damping. We therefore recommend **R-iKFAD0** ( $\gamma = 0$ ) as the practical default since it has one fewer hyperparameter.

#### 4.3 Robustness to Hyperparameters

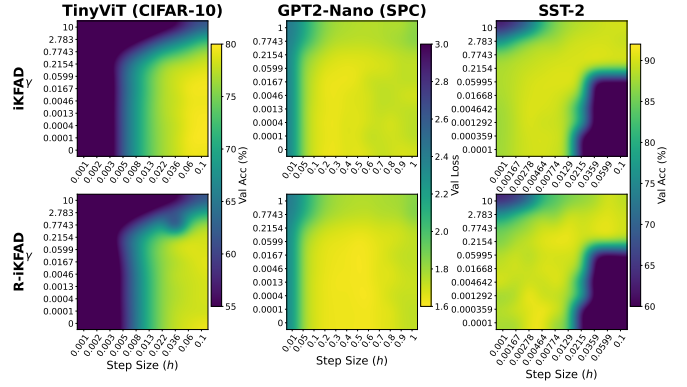
Figure 2 shows  $\gamma$ - $h$  grid sweeps for R-iKFAD and iKFAD, revealing comparably broad high-performance plateaus. Practically, this allows for coarse hyperparameter tuning. This robustness stems from the adaptive friction mechanism: momentum spikes from large step sizes  $h$  automatically scale up the rank-1 factors  $R$  and  $C$ , increasing  $\tilde{\xi}$  to dampen

Table 2: Lowest test loss achieved during training (mean $\pm$ std over 10 seeds; GPT2-S: single seed due to computational limitations). Lower is better. R-iKFAD0 matches or improves upon iKFAD0 across all benchmarks while using  $\approx 2\times$  less optimizer memory.

| Dataset    | Adam                                | Adafactor-m       | iKFAD0            | R-iKFAD0 (ours)                     |
|------------|-------------------------------------|-------------------|-------------------|-------------------------------------|
| GPT2-Nano  | $1.643 \pm 0.008$                   | $1.661 \pm 0.007$ | $1.640 \pm 0.008$ | <b><math>1.633 \pm 0.011</math></b> |
| TinyViT    | <b><math>0.593 \pm 0.015</math></b> | $0.596 \pm 0.010$ | $0.606 \pm 0.011$ | $0.606 \pm 0.012$                   |
| DistilBERT | $0.292 \pm 0.005$                   | $0.280 \pm 0.007$ | $0.269 \pm 0.005$ | <b><math>0.268 \pm 0.003</math></b> |
| GPT2-S     | 3.301                               | 3.287             | 3.263             | <b>3.242</b>                        |



(a) Accuracy averaged over 10 seeds. R-iKFAD0 matches iKFAD0 and Adam. Error bars denote one std.



(b) Two-dimensional  $\gamma$ - $h$  sensitivity for R-iKFAD (top) and iKFAD (bottom). Each cell shows best performance at a fixed  $(h, \gamma)$  grid point; brighter = better. R-iKFAD retains comparably broad high-performance plateaus.

Figure 2: Accuracy results and  $\gamma$ - $h$  robustness sweeps for R-iKFAD0 and baselines.

oscillations. This self-stabilizing feedback prevents overshooting, confirming that R-iKFAD inherits iKFAD’s tolerance to aggressive hyperparameters [7] without degradation from rank-1 compression.

## 5 Conclusion

We introduced **R-iKFAD**, an optimizer that applies rank-1 compression to iKFAD’s friction tensor to reduce per-layer friction storage from  $\mathcal{O}(mn)$  to  $\mathcal{O}(m+n)$ . By maintaining a full momentum buffer alongside low-rank friction vectors, R-iKFAD occupies a sublinear-friction regime derived from provably convergent friction dynamics. The goal is not to surpass Adam, but to preserve iKFAD’s established performance parity with Adam while cutting auxiliary optimizer memory in half. Our experiments across GPT2-Nano, TinyViT, DistilBERT, and GPT2-S confirm that R-iKFAD matches iKFAD in convergence speed, best achieved accuracy, and hyperparameter robustness at half the auxiliary memory cost. On the theoretical side, we analyze the continuous-time dynamics in both damping regimes: exponential convergence for  $\gamma > 0$  under strong convexity, and for the undamped case  $\gamma = 0$  that we actually deploy, unconditional convergence together with algebraic rates  $\mathcal{O}(t^{-1})$  and  $\mathcal{O}(t^{-1/2})$  separated by the regularization scale. To our knowledge this is the first convergence rate for a rank-1 factored optimizer in continuous time, and the first such analysis that does not require positive damping. Discrete-time analyses for  $\gamma > 0$ , for  $\gamma = 0$ , and for the stochastic-gradient setting are in progress and will appear in the extended version of this work; validating performance at larger scale remains future work.

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## A Update Rules for Splitting Discretization

To derive a practical discrete-time algorithm from the continuous-time dynamics (11)–(14), we use an operator-splitting approach [12]. The full vector field is decomposed into four sub-operators whose individual flows can be integrated analytically:

$$\begin{pmatrix} \dot{X} \\ \dot{P} \\ \dot{R} \\ \dot{C} \end{pmatrix} = \underbrace{\begin{pmatrix} P \\ 0 \\ 0 \\ 0 \end{pmatrix}}_A + \underbrace{\begin{pmatrix} 0 \\ -\nabla f(X) \\ 0 \\ 0 \end{pmatrix}}_B + \underbrace{\begin{pmatrix} 0 \\ -\frac{R C^\top}{\mathbf{1}_m^\top R + \epsilon_{\text{stab}}} \odot P \\ \frac{1}{\mu} [P]^2 \mathbf{1}_n - \alpha R \\ \frac{1}{\mu} \mathbf{1}_m^\top [P]^2 - \alpha C \end{pmatrix}}_C + \underbrace{\begin{pmatrix} 0 \\ -\gamma P \\ 0 \\ 0 \end{pmatrix}}_D. \quad (16)$$

Sub-flows A, B, and D integrate analytically: A gives position drift; B applies the gradient kick; D applies linear friction (becoming the identity when  $\gamma = 0$ , i.e., R-iKFAD0). Sub-flow C couples momentum  $P$  and friction factors  $(R, C)$  and is integrated semi-analytically via an internal symmetric freeze-then-update scheme. The composition applied in practice is a first-order Lie–Trotter splitting:

$$\Phi_h = \Phi_h^D \circ \Phi_h^C \circ \Phi_h^A \circ \Phi_h^B,$$

where sub-steps are executed in the sequence  $B \rightarrow A \rightarrow C \rightarrow D$  (BACD) [7]. Position updates use the gradient-kicked momentum  $P^{(B)}$  rather than  $P_n$ . Table 3 summarizes the explicit analytical update rules for each sub-step across the complete BACD sequence over time step  $h$ .

Table 3: Analytical sub-step update rules for the BACD splitting sequence over time step  $h$ .

| Sub-step | Update Equation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>B</b> | $P^{(B)} = P_n - h \nabla f(X_n)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>A</b> | $X_{n+1} = X_n + h P^{(B)}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>C</b> | $\tilde{\xi}_n = \frac{R_n C_n^\top}{\mathbf{1}_m^\top R_n + \epsilon_{\text{stab}}}$ $P^{(1)} = P^{(B)} \odot \exp\left(-\frac{h}{2} \tilde{\xi}_n\right)$ $R_{n+1} = e^{-\alpha h} R_n + \frac{1 - e^{-\alpha h}}{\mu \alpha} [P^{(1)}]^2 \mathbf{1}_n$ $C_{n+1} = e^{-\alpha h} C_n + \frac{1 - e^{-\alpha h}}{\mu \alpha} \mathbf{1}_m^\top [P^{(1)}]^2$ $\tilde{\xi}_{n+1} = \frac{R_{n+1} C_{n+1}^\top}{\mathbf{1}_m^\top R_{n+1} + \epsilon_{\text{stab}}}$ $P^{(2)} = P^{(1)} \odot \exp\left(-\frac{h}{2} \tilde{\xi}_{n+1}\right)$ |
| <b>D</b> | $P_{n+1} = P^{(2)} \odot e^{-\gamma h} \quad (\text{identity when } \gamma = 0)$                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

Within sub-flow C, no closed-form joint solution exists for coupled  $P$  and  $(R, C)$ . We apply a symmetric half-damp sequence: evaluating  $\exp(-\frac{h}{2} \tilde{\xi})$  twice around an exact exponential ODE factor update improves integration accuracy to  $\mathcal{O}(h^2)$  within sub-flow C alone. Using the post-first-half-damp momentum  $P^{(1)}$  for factor updates reduces forcing magnitude and prevents over-accumulation.

## B Optimal Hyperparameters

Across models and tasks, the best-performing scales for  $\mu$  (iKFAD0, R-iKFAD0) varied by several orders of magnitude. This variation is likely related to the scale of the momentum variables during training, since  $\mu$  controls the strength of the adaptive friction terms. Normalizing the momentum variables across iterations could potentially reduce this sensitivity, but we leave this for future work.

Hyperparameters were selected by minimum validation loss during the Optuna sweeps; all reported results are computed on held-out test data.

Experimental configurations:

- **TinyViT – CIFAR-10:** batch size 128, 25 epochs, 100 trials.  $h \in [10^{-6}, 10^{-1}]$ ,  $\alpha \in [10^{-3}, 1]$ ,  $\mu \in [10^{-8}, 10]$  for iKFAD0 and R-iKFAD0.
- **DistilBERT – SST-2:** batch size 16, 2 epochs, 80 trials.  $h \in [10^{-6}, 10^{-1}]$ ,  $\alpha \in [10^{-3}, 1]$ ,  $\mu \in [10^{-8}, 10]$  for iKFAD0 and R-iKFAD0.
- **GPT2-Nano – Shakespeare:** batch size 16, 5000 steps for the hyperparameter sweeps and final reported runs, 500 trials.  $h \in [10^{-6}, 5 \times 10^{-1}]$ ,  $\alpha \in [10^{-5}, 10]$ ,  $\mu \in [10^{-8}, 10]$  for iKFAD0 and R-iKFAD0.
- **GPT2-S – OpenWebText:** batch size 16, 75001 steps for the hyperparameter sweeps and 200001 steps for the final reported runs, 80 trials.  $h \in [10^{-6}, 10^{-1}]$ ,  $\alpha \in [10^{-3}, 1]$ ,  $\mu \in [10^{-8}, 10]$  for iKFAD0 and  $\mu \in [10^{-9}, 10^{-2}]$  for R-iKFAD0.

Table 4: Optimized hyperparameters for all experiments. All iKFAD0 and R-iKFAD0 entries have  $\gamma = 0$ . Dashes indicate parameters not used by the method.

|                                | $h$                     | $\alpha$ | $\mu$                   | $\beta_1$ | $\beta_2$ |
|--------------------------------|-------------------------|----------|-------------------------|-----------|-----------|
| <b>TinyViT (CIFAR-10)</b>      |                         |          |                         |           |           |
| iKFAD0                         | 0.0725                  | 0.0899   | $1.2643 \times 10^{-5}$ | –         | –         |
| R-iKFAD0                       | 0.0860                  | 0.5327   | $9.4765 \times 10^{-7}$ | –         | –         |
| Adam                           | $5.5311 \times 10^{-4}$ | –        | –                       | 0.8608    | 0.8852    |
| Adafactor-m                    | $5.6050 \times 10^{-4}$ | –        | –                       | 0.8907    | –         |
| <b>DistilBERT (SST-2)</b>      |                         |          |                         |           |           |
| iKFAD0                         | 0.0165                  | 0.0574   | $9.8948 \times 10^{-6}$ | –         | –         |
| R-iKFAD0                       | 0.0423                  | 0.0060   | $8.1874 \times 10^{-7}$ | –         | –         |
| Adam                           | $3.82 \times 10^{-5}$   | –        | –                       | 0.9187    | 0.9825    |
| Adafactor-m                    | $2.5638 \times 10^{-5}$ | –        | –                       | 0.9622    | –         |
| <b>GPT2-Nano (Shakespeare)</b> |                         |          |                         |           |           |
| iKFAD0                         | 0.4941                  | 2.1955   | $4.5521 \times 10^{-6}$ | –         | –         |
| R-iKFAD0                       | 0.4842                  | 2.8475   | $1.9407 \times 10^{-6}$ | –         | –         |
| Adam                           | $1.6803 \times 10^{-3}$ | –        | –                       | 0.8876    | 0.9265    |
| Adafactor-m                    | $2.9923 \times 10^{-3}$ | –        | –                       | 0.8893    | –         |
| <b>GPT2-S (OpenWebText)</b>    |                         |          |                         |           |           |
| iKFAD0                         | 0.0938                  | 0.1933   | $4.2726 \times 10^{-7}$ | –         | –         |
| R-iKFAD0                       | 0.0918                  | 0.0579   | $1.3637 \times 10^{-6}$ | –         | –         |
| Adam                           | $1.0478 \times 10^{-4}$ | –        | –                       | 0.9000    | 0.9971    |
| Adafactor-m                    | $7.2777 \times 10^{-4}$ | –        | –                       | 0.8865    | –         |

For all experiments, Adam was swept over  $h \in [10^{-6}, 10^{-2}]$ ,  $\beta_1 \in [0.85, 0.999]$ ,  $\beta_2 \in [0.85, 0.999]$ . Adafactor-m was swept over  $h \in [10^{-5}, 10^{-2}]$  and  $\beta_1 \in [0.85, 0.999]$ , with no relative step or warmup. Adafactor-m has no tunable  $\beta_2$ : as originally proposed [17] and in standard practice, the second-moment decay follows the prescribed schedule  $\beta_{2,t} = 1 - t^{-0.8}$  (`decay_rate` = `-0.8`), which is why its  $\beta_2$  entries are left blank. We enable  $\beta_1 > 0$  rather than the conventional momentum-free setting so that Adafactor-m occupies the same auxiliary-memory regime as R-iKFAD (Table 5). All sweeps used Optuna’s [1] TPE sampler. Table 4 reports the best hyperparameters found. iKFAD0 and R-iKFAD0 are the  $\gamma = 0$  variants; all such entries satisfy  $\gamma = 0$  by definition. For iKFAD0 and R-iKFAD0,  $h$  denotes the step size and  $\mu$  the adaptive damping coefficient. For Adam and Adafactor-m,  $h$  is the learning rate.

## C Ablation: $\gamma = 0$ versus $\gamma > 0$

Figure 3 compares R-iKFAD ( $\gamma > 0$ , with linear damping) against R-iKFAD0 ( $\gamma = 0$ , without linear damping) on GPT2-Nano, TinyViT, and DistilBERT (SST-2). Adding the  $\gamma$  term yields no consistent improvement over the  $\gamma = 0$  default on any benchmark, confirming that the rank-1 friction encoded in  $R$  and  $C$  provides sufficient adaptive damping. This is consistent with the analogous finding for iKFAD in [7]. For the  $\gamma > 0$  variant,  $\gamma$  was not fixed by hand: we used the same settings as R-iKFAD0 given in Appendix B and additionally swept  $\gamma$  uniformly over  $[10^{-6}, 10]$ .



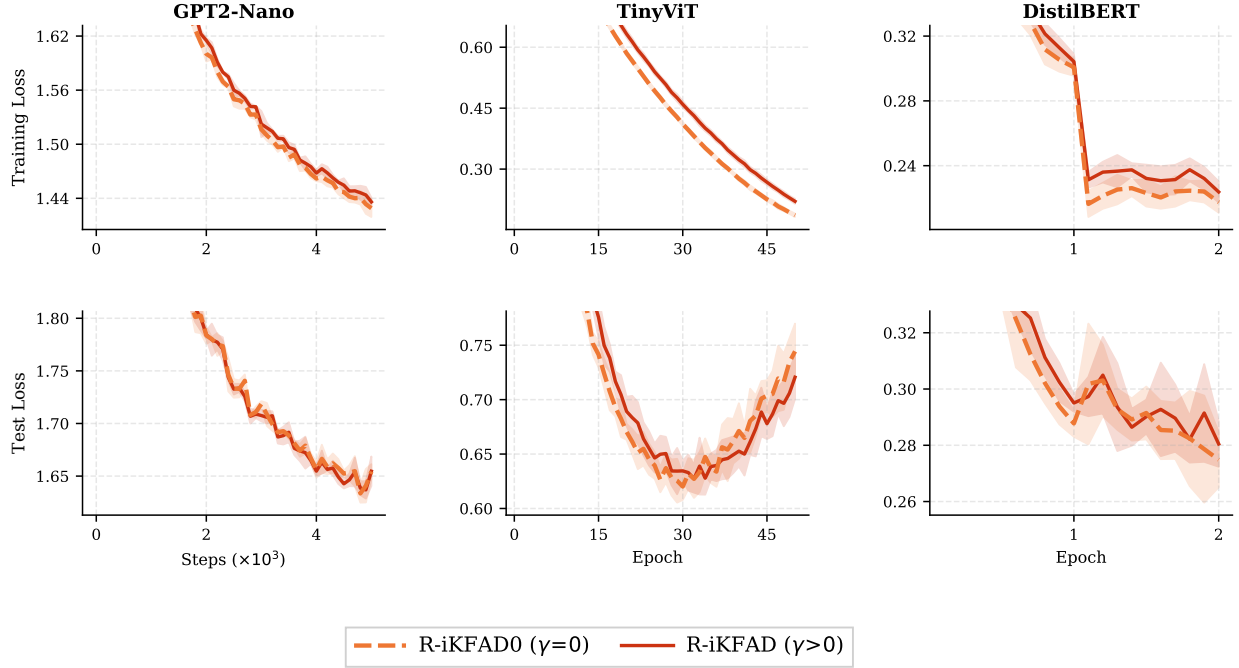


Figure 3: Training and test loss for R-iKFAD ( $\gamma > 0$ ) vs. R-iKFAD0 ( $\gamma = 0$ ) on GPT2-Nano, TinyViT, and DistilBERT (SST-2), averaged over 10 seeds (shaded = std). The two variants are indistinguishable in performance, supporting R-iKFAD0 as the recommended default.

## D Memory Analysis

Table 5: Optimizer auxiliary state (MiB, float32), computed over all model parameters. R-iKFAD achieves approximately  $2\times$  savings over both iKFAD and Adam because the rank-1 friction factors ( $R_\ell, C_\ell$ ) are negligible relative to the full momentum buffer. The Adafactor-m baseline used throughout enables its optional first moment ( $\beta_1 > 0$ ; Section 4) and therefore carries an auxiliary footprint identical to R-iKFAD’s, making the two directly comparable at matched memory. Momentum-free Adafactor is listed for reference.

| Model      | $N$    | Adam  | Adafactor-m | Adafactor | iKFAD | R-iKFAD               |
|------------|--------|-------|-------------|-----------|-------|-----------------------|
| TinyViT    | 5.0M   | 30.7  | 15.6        | 0.2       | 30.7  | 15.6 (2.0 $\times$ )  |
| DistilBERT | 80.0M  | 510.8 | 256.1       | 0.7       | 510.8 | 256.1 (2.0 $\times$ ) |
| GPT2-Nano  | 0.8M   | 6.1   | 3.1         | < 0.1     | 6.1   | 3.1 (2.0 $\times$ )   |
| GPT2-S     | 124.4M | 948.9 | 475.4       | 0.9       | 948.9 | 475.4 (2.0 $\times$ ) |

## E Rank-1 iKFAD Convergence Analysis

We analyze the continuous-time Rank-1 iKFAD dynamics for a matrix parameter  $X \in \mathbb{R}^{m \times n}$  (the vector case follows by flattening). The system is given by

$$\dot{X} = P, \quad (17a)$$

$$\dot{P} = -\nabla f(X) - \tilde{\xi} \odot P - \gamma P, \quad (17b)$$

$$\dot{R} = \frac{1}{\mu} [P]^2 \mathbf{1}_n - \alpha R, \quad (17c)$$

$$\dot{C} = \frac{1}{\mu} \mathbf{1}_m^\top [P]^2 - \alpha C, \quad (17d)$$

where  $P \in \mathbb{R}^{m \times n}$ ,  $R \in \mathbb{R}^m$ ,  $C \in \mathbb{R}^n$ , and  $\gamma \geq 0$ ,  $\alpha, \mu > 0$  are hyperparameters. Boundedness (Lemma 1) holds for  $\gamma \geq 0$ ; the exponential convergence results below additionally require  $\gamma > 0$  (see Remark 2). The rank-1 friction tensor

$\tilde{\xi} \in \mathbb{R}^{m \times n}$  is defined element-wise as

$$\tilde{\xi}_{ij} = \frac{R_i C_j}{\sum_{k=1}^m R_k + \epsilon_{\text{stab}}},$$

with  $\epsilon_{\text{stab}} > 0$  a small constant preventing division by zero. The notation  $\odot$  denotes element-wise multiplication,  $[\cdot]^2$  element-wise squaring, and  $\langle \cdot, \cdot \rangle_F$  the Frobenius inner product.

### E.1 Dynamical properties and boundedness

Throughout this appendix  $X^*$  denotes a global minimizer of  $f$ . Under the hypotheses used below such a point exists, because  $f$  is continuous and coercive, and it satisfies

$$\nabla f(X^*) = 0 \quad \text{and} \quad f(X) \geq f(X^*) \quad \text{for all } X. \quad (18)$$

Both properties are used repeatedly and neither follows from the growth condition (21) alone: expanding (21) along  $X = X^* + tu$  at order  $t$  gives  $(1-a)\langle u, \nabla f(X^*) \rangle_F \geq 0$  for every  $u$ , which forces  $\nabla f(X^*) = 0$  only when  $a \neq 1$ ; since  $a = 1$  is admissible under strong convexity (Remark 1), the property must be imposed rather than derived.

**Lemma 1** (Boundedness of solutions). *Assume  $f \in C^2$  with  $\nabla f$  locally Lipschitz,  $f(X) \rightarrow +\infty$  as  $\|X\|_F \rightarrow +\infty$ , and let  $X^*$  be a global minimizer of  $f$  as in (18). Then for any initial condition  $(X_0, P_0, R_0, C_0)$  with  $R_0, C_0 \geq 0$  entrywise, the solution of (17a)–(17d) is well-defined for all  $t \geq 0$ , and there exists a compact set  $\mathcal{K}$  containing the trajectory. In particular, with*

$$P_{\max}^2 := 2(f(X_0) - f(X^*) + \tfrac{1}{2}\|P_0\|_F^2), \quad R_{\max} := \|R_0\| + \frac{P_{\max}^2}{\alpha\mu}, \quad C_{\max} := \|C_0\| + \frac{P_{\max}^2}{\alpha\mu},$$

all of which are finite and independent of  $t$ , we have for all  $t \geq 0$

$$0 \leq R_i(t) \leq R_{\max}, \quad 0 \leq C_j(t) \leq C_{\max}, \quad \|P(t)\|_F \leq P_{\max},$$

and moreover the friction tensor obeys the bound

$$0 \leq \tilde{\xi}_{ij}(t) \leq \xi_{\max} := \min\left\{C_{\max}, \frac{R_{\max}C_{\max}}{\epsilon_{\text{stab}}}\right\}. \quad (19)$$

Note that  $P_{\max}$  depends on neither  $\mu$  nor  $\gamma$ .

*Proof.* The ODEs (17c)–(17d) have non-negative forcing and linear decay, so  $R_0, C_0 \geq 0$  implies  $R_i(t), C_j(t) \geq 0$  for all  $t \geq 0$ , and hence  $\tilde{\xi}_{ij} \geq 0$  throughout.

Consider the reduced energy

$$\mathcal{E}(X, P) = f(X) - f(X^*) + \tfrac{1}{2}\|P\|_F^2,$$

which omits the  $R, C$  contributions. Differentiating along trajectories and canceling the gradient terms,

$$\dot{\mathcal{E}} = \langle \nabla f(X), P \rangle_F + \langle P, -\nabla f(X) - \tilde{\xi} \odot P - \gamma P \rangle_F = -\gamma\|P\|_F^2 - \sum_{i,j} \tilde{\xi}_{ij} P_{ij}^2 \leq 0, \quad (20)$$

both terms being non-positive because  $\gamma \geq 0$  and  $\tilde{\xi} \geq 0$  entrywise. No bound on  $\|P\|_F$  is used in deriving (20), so the estimate is not circular. Consequently  $\mathcal{E}(t) \leq \mathcal{E}(0)$  for all  $t$ , which yields at once

$$\|P(t)\|_F^2 \leq 2\mathcal{E}(0) =: P_{\max}^2,$$

and, since  $f(X) - f(X^*) \leq \mathcal{E}(0)$  with  $f$  coercive, confinement of  $X(t)$  to a compact sublevel set of  $f$ .

With  $\|P\|_F \leq P_{\max}$  now established independently, the factors follow from the explicit linear filter. Integrating (17c),

$$R(t) = e^{-\alpha t} R_0 + \frac{1}{\mu} \int_0^t e^{-\alpha(t-s)} [P(s)]^2 \mathbf{1}_n \, ds,$$

and using the sharp, dimension-free estimate  $\|[P]^2 \mathbf{1}_n\| \leq \|P\|_F^2 \leq P_{\max}^2$  (the entries of  $[P]^2 \mathbf{1}_n$  are non-negative and sum to  $\|P\|_F^2$ , so its 2-norm is at most its 1-norm),

$$\|R(t)\| \leq \|R_0\| + \frac{P_{\max}^2}{\alpha\mu} =: R_{\max},$$

with the analogous bound  $C_{\max} := \|C_0\| + P_{\max}^2/(\alpha\mu)$ . All four variables therefore remain in a compact set  $\mathcal{K}$  for all  $t \geq 0$ , and global existence follows from the standard continuation criterion for locally Lipschitz vector fields.

Finally we establish (19). Two bounds are available. Bounding the denominator below by  $\epsilon_{\text{stab}}$  gives  $\tilde{\xi}_{ij} \leq R_{\max} C_{\max}/\epsilon_{\text{stab}}$ , which diverges as  $\epsilon_{\text{stab}} \downarrow 0$ . The second is sharper and uniform in  $\epsilon_{\text{stab}}$ : since every  $R_k \geq 0$  we have  $R_i \leq \mathbf{1}_m^\top R$ , hence

$$\tilde{\xi}_{ij} = \frac{R_i}{\mathbf{1}_m^\top R + \epsilon_{\text{stab}}} C_j \leq C_j \leq C_{\max},$$

and (19) is the minimum of the two. The constant  $\epsilon_{\text{stab}} > 0$  remains necessary for well-posedness at  $R = 0$ , which is exactly how the implementation initializes the factors, but it does not enter the bound on  $\tilde{\xi}$ , and hence does not enter the convergence rate.  $\square$

## E.2 Exponential convergence of the continuous dynamics

The rank-1 factorization is not free in this regime. Because  $\tilde{\xi}_{ij} \leq C_j$ , the factor dynamics inject energy into the Lyapunov function cubically in the state while the available negative terms are quadratic, and the mismatch must be paid for by down-weighting the auxiliary block; in full-friction iKFAD the corresponding terms cancel exactly. Remark 3 makes this precise, and it is the sense in which the condition below is the price of compression rather than an artifact of the argument.

**Theorem 2** (Exponential convergence of Rank-1 iKFAD). *Assume  $f \in C^2$  satisfies the strong convexity condition: there exist  $a, b > 0$  such that for all  $X$ ,*

$$\langle X - X^*, \nabla f(X) \rangle_F \geq a(f(X) - f(X^*)) + b\|X - X^*\|_F^2. \quad (21)$$

where  $X^*$  is a global minimizer of  $f$  as in (18). Let  $\gamma > 0$  and  $\alpha, \mu, \epsilon_{\text{stab}} > 0$ , and fix an initial condition  $(X_0, P_0, R_0, C_0)$  with  $R_0, C_0 \geq 0$  entrywise. Then there exist constants  $\kappa > 0$  and  $C_* > 0$  such that the solution of (17a)–(17d) satisfies for all  $t \geq 0$ ,

$$f(X(t)) - f(X^*) + \|X(t) - X^*\|_F^2 + \|P(t)\|_F^2 + \|R(t)\|^2 + \|C(t)\|^2 \leq C_* e^{-\kappa t}.$$

**Remark 1** (Coercivity is implied, not assumed). *Lemma 1 requires  $f$  to be coercive, which Theorem 2 does not assume separately: it follows from (21). Since  $\nabla f(X^*) = 0$  by (18), expanding (21) along  $X = X^* + tu$  at second order gives  $(1 - \frac{a}{2})\langle u, \nabla^2 f(X^*)u \rangle_F \geq b\|u\|_F^2 > 0$ , so  $a < 2$ . Writing  $\phi(t) := f(X^* + tu) - f(X^*)$ , (21) reads  $t\phi'(t) \geq a\phi(t) + bt^2\|u\|_F^2$ , i.e.  $(\phi(t)/t^a)' \geq bt^{1-a}\|u\|_F^2$ , which integrates to*

$$f(X) - f(X^*) \geq \frac{b}{2-a} \|X - X^*\|_F^2.$$

This supplies both the coercivity needed by Lemma 1 and the inequality  $f \geq f(X^*)$  of (18), and it is what makes  $P_{\max}^2 = 2\mathcal{E}(0)$  non-negative and finite.

**Remark 2** (On the restriction  $\gamma > 0$ ). *The hypothesis  $\gamma > 0$  is not an artifact of the proof. At  $\gamma = 0$  the linearization of (17a)–(17d) about  $(X^*, 0, 0, 0)$  is*

$$\dot{X} = P, \quad \dot{P} = -\nabla^2 f(X^*)(X - X^*), \quad \dot{R} = -\alpha R, \quad \dot{C} = -\alpha C,$$

since  $R$  and  $C$  are driven by  $[P]^2$  and  $\tilde{\xi}$  is a ratio in which they appear, the friction term  $\tilde{\xi} \odot P$  vanishes to higher order at the equilibrium in either regime of the denominator. Consequently the  $(X, P)$  block reduces to  $\begin{bmatrix} 0 & I \\ -H & 0 \end{bmatrix}$  with  $H = \nabla^2 f(X^*) \succeq 0$ , which has zero trace and spectrum symmetric under  $\lambda \mapsto -\lambda$ , so it is not Hurwitz for any  $H$ . For a  $C^1$  vector field a non-Hurwitz linearization precludes local exponential stability [9, Theorem 4.15]; the equilibrium is therefore not exponentially stable at  $\gamma = 0$ , and no bound of the form asserted in Theorem 2 can hold there.

The Lyapunov argument below also fails at  $\gamma = 0$  for a separate and more elementary reason: the term  $-\gamma\|P\|_F^2$  is the only negative contribution available to absorb the energy that the factor dynamics inject (Remark 3), and it is absent. What replaces exponential decay is therefore not a defect of this argument but a genuinely different rate, and it is established in Appendix E.3: convergence holds unconditionally (Theorem 3), at the algebraic rate  $\mathcal{O}(t^{-1})$  for  $\epsilon_{\text{stab}} = 0$  and  $\mathcal{O}(t^{-1/2})$  for  $\epsilon_{\text{stab}} > 0$  (Theorem 8). Since the experiments in Section 4 use  $\gamma = 0$ , that is the regime the deployed method occupies.

**Remark 3** (The auxiliary channel is a genuine energy source). *It is worth being explicit about why the weight  $B$  introduced in (22) is needed at all, since this is precisely the price of rank-1 compression. Before any weighting, the friction and factor terms of (25) combine to a non-negative quantity: by (19),  $\tilde{\xi}_{ij} \leq C_j$ , hence*

$$-\langle P, \tilde{\xi} \odot P \rangle_F + \langle R, [P]^2 \mathbf{1}_n \rangle + \langle C, \mathbf{1}_m^\top [P]^2 \rangle = \sum_{i,j} P_{ij}^2 (R_i + C_j - \tilde{\xi}_{ij}) \geq \sum_{i,j} R_i P_{ij}^2 \geq 0$$

identically: the factor dynamics inject energy into  $\mathcal{W}_\epsilon$ , cubically in the state, whereas the available negative terms are quadratic. In full-friction iKFAD the corresponding terms cancel exactly, because there the friction obeys  $\dot{\xi} = [P]^2/\mu - \alpha\xi$  elementwise and  $\langle P, \xi \odot P \rangle_F$  is matched term for term by  $\langle \xi, [P]^2 \rangle$ . Under the rank-1 factorization no such cancellation is available, and the mismatch must be paid for, here by down-weighting the auxiliary block. No sharpening of  $P_{\max}$  or of (26) can remove it.

*Proof.* We employ the perturbed Lyapunov function

$$\mathcal{W}_\epsilon(X, P, R, C) = f(X) - f(X^*) + \frac{1}{2}\|P\|_F^2 + \frac{\mu B}{2}(\|R\|^2 + \|C\|^2) + \epsilon\langle X - X^*, P \rangle_F + \frac{\epsilon}{2}\|X - X^*\|_F^2, \quad (22)$$

where  $\epsilon \in (0, \frac{1}{4}]$  is a small parameter to be chosen later. Applying Young's inequality in the form  $\epsilon|\langle X - X^*, P \rangle_F| \leq \epsilon(\frac{1}{4}\|X - X^*\|_F^2 + \|P\|_F^2)$  and using  $\epsilon \leq 1/4$  gives

$$\mathcal{W}_\epsilon \geq f(X) - f(X^*) + \frac{1}{4}\|P\|_F^2 + \frac{\mu B}{2}(\|R\|^2 + \|C\|^2) + \frac{\epsilon}{4}\|X - X^*\|_F^2, \quad (23)$$

$$\mathcal{W}_\epsilon \leq f(X) - f(X^*) + \frac{3}{4}\|P\|_F^2 + \frac{\mu B}{2}(\|R\|^2 + \|C\|^2) + \frac{3\epsilon}{2}\|X - X^*\|_F^2. \quad (24)$$

Differentiating  $\mathcal{W}_\epsilon$  along trajectories of (17a)–(17d) yields

$$\begin{aligned} \dot{\mathcal{W}}_\epsilon &= \langle \nabla f(X), P \rangle_F + \langle P, -\nabla f(X) - \tilde{\xi} \odot P - \gamma P \rangle_F \\ &\quad + \mu B \langle R, \dot{R} \rangle + \mu B \langle C, \dot{C} \rangle \\ &\quad + \epsilon\|P\|_F^2 - \epsilon\langle X - X^*, \nabla f(X) \rangle_F - \epsilon\langle X - X^*, \tilde{\xi} \odot P \rangle_F + \epsilon(1 - \gamma)\langle X - X^*, P \rangle_F. \end{aligned}$$

Canceling  $\langle \nabla f(X), P \rangle_F$  with  $-\langle P, \nabla f(X) \rangle_F$  and substituting the auxiliary dynamics gives

$$\begin{aligned} \dot{\mathcal{W}}_\epsilon &= -\gamma\|P\|_F^2 - \langle P, \tilde{\xi} \odot P \rangle_F + B\langle R, [P]^2 \mathbf{1}_n \rangle + B\langle C, \mathbf{1}_m^\top [P]^2 \rangle - B\alpha\mu(\|R\|^2 + \|C\|^2) \\ &\quad + \epsilon\|P\|_F^2 - \epsilon\langle X - X^*, \nabla f(X) \rangle_F - \epsilon\langle X - X^*, \tilde{\xi} \odot P \rangle_F + \epsilon(1 - \gamma)\langle X - X^*, P \rangle_F. \end{aligned} \quad (25)$$

Define  $\mathcal{M} := -\langle P, \tilde{\xi} \odot P \rangle_F + B\langle R, [P]^2 \mathbf{1}_n \rangle + B\langle C, \mathbf{1}_m^\top [P]^2 \rangle$ , the collection of all terms in (25) generated by the friction and factor dynamics. Expanding element-wise,

$$\mathcal{M} = \sum_{i,j} P_{ij}^2 \left( BR_i + BC_j - \frac{R_i C_j}{\sum_k R_k + \epsilon_{\text{stab}}} \right).$$

Since  $-\langle P, \tilde{\xi} \odot P \rangle_F \leq 0$ , we can drop this negative contribution and bound the positive terms. The vector  $[P]^2 \mathbf{1}_n$  is entrywise non-negative with entries summing to  $\|P\|_F^2$ , so

$$\|[P]^2 \mathbf{1}_n\|_2 \leq \|[P]^2 \mathbf{1}_n\|_1 = \|P\|_F^2,$$

and likewise  $\|\mathbf{1}_m^\top [P]^2\|_2 \leq \|P\|_F^2$ . This estimate is sharp, with equality when a single row (respectively column) of  $P$  is non-zero, and it carries no dimensional factor. Hence by Cauchy–Schwarz followed by Young's inequality with any  $\theta > 0$ , using  $\|P\|_F \leq P_{\max}$  from Lemma 1,

$$\begin{aligned} \langle R, [P]^2 \mathbf{1}_n \rangle &\leq \|R\| \|P\|_F^2 \leq \frac{\theta}{2}\|R\|^2 + \frac{P_{\max}^2}{2\theta}\|P\|_F^2, \\ \langle C, \mathbf{1}_m^\top [P]^2 \rangle &\leq \|C\| \|P\|_F^2 \leq \frac{\theta}{2}\|C\|^2 + \frac{P_{\max}^2}{2\theta}\|P\|_F^2. \end{aligned}$$

Multiplying by  $B$  and adding, and discarding  $-\langle P, \tilde{\xi} \odot P \rangle_F \leq 0$ ,

$$\mathcal{M} \leq \frac{B\theta}{2}(\|R\|^2 + \|C\|^2) + \frac{B P_{\max}^2}{\theta}\|P\|_F^2. \quad (26)$$

For the cross-terms involving  $X - X^*$ , we apply Young's inequality with constants  $\eta, \delta > 0$ :

$$\epsilon(1 - \gamma)\langle X - X^*, P \rangle_F \leq \epsilon|1 - \gamma|\left(\eta\|X - X^*\|_F^2 + \frac{1}{4\eta}\|P\|_F^2\right), \quad (27)$$

$$-\epsilon\langle X - X^*, \tilde{\xi} \odot P \rangle_F \leq \epsilon\left(\delta\|X - X^*\|_F^2 + \frac{\xi_{\max}^2}{4\delta}\|P\|_F^2\right), \quad (28)$$

where  $\xi_{\max}$  is the entry-wise bound on  $\tilde{\xi}$  established in (19). We use it in the form  $\xi_{\max} \leq C_{\max}$ , which is uniform in  $\epsilon_{\text{stab}}$ ; the alternative branch  $R_{\max}C_{\max}/\epsilon_{\text{stab}}$  is also valid but diverges as  $\epsilon_{\text{stab}} \downarrow 0$  and is never needed. Consequently  $\epsilon_{\text{stab}}$  enters the hypotheses only through well-posedness and does not appear in  $\kappa$ .

Substituting (26), (27), (28) and the strong convexity condition (21) into (25) yields

$$\begin{aligned} \dot{\mathcal{W}}_{\epsilon} \leq & -\left[\gamma - \epsilon - \frac{BP_{\max}^2}{\theta} - \epsilon\left(\frac{|1-\gamma|}{4\eta} + \frac{\xi_{\max}^2}{4\delta}\right)\right]\|P\|_F^2 \\ & - B\left(\alpha\mu - \frac{\theta}{2}\right)\|R\|^2 - B\left(\alpha\mu - \frac{\theta}{2}\right)\|C\|^2 \\ & - a\epsilon(f(X) - f(X^*)) - \epsilon\left[b - \eta|1-\gamma| - \delta\right]\|X - X^*\|_F^2. \end{aligned} \quad (29)$$

We now choose the free parameters to make all coefficients negative. Set  $\theta = \alpha\mu$ , so that  $B(\alpha\mu - \frac{\theta}{2}) = \frac{B\alpha\mu}{2} > 0$  for every  $B > 0$ , and likewise for  $C$ ; the auxiliary coefficients are therefore positive irrespective of  $B$ , and the relative decay rate  $\frac{B\alpha\mu/2}{\mu B/2} = \alpha$  does not depend on  $B$  at all. The leak into the momentum coefficient is  $K_1 := BP_{\max}^2/\theta = BP_{\max}^2/(\alpha\mu)$ . Next choose

$$\eta = \frac{b}{4(1+|1-\gamma|)}, \quad \delta = \frac{b}{4},$$

which is well defined for every  $\gamma \geq 0$ , including  $\gamma = 1$ . Since  $|1-\gamma|/(1+|1-\gamma|) < 1$  we have  $\eta|1-\gamma| < b/4$ , hence  $b - \eta|1-\gamma| - \delta > b/2 > 0$ . Substituting  $\eta$  into (27) gives  $|1-\gamma|/(4\eta) = |1-\gamma|(1+|1-\gamma|)/b$ , so the coefficient of  $\|P\|_F^2$  becomes

$$\Gamma(\epsilon) := \gamma - K_1 - \epsilon\Lambda, \quad \Lambda := 1 + \frac{|1-\gamma|(1+|1-\gamma|) + \xi_{\max}^2}{b}.$$

Since  $K_1 = BP_{\max}^2/(\alpha\mu)$  is proportional to  $B$ , and  $B$  is a free parameter of the proof which appears nowhere in the statement of Theorem 2, we may simply set

$$B := \min\left\{1, \frac{\gamma\alpha\mu}{2P_{\max}^2}\right\}, \quad (30)$$

which gives  $K_1 \leq \gamma/2$  and hence  $\gamma - K_1 \geq \gamma/2 > 0$  for every  $\gamma, \alpha, \mu > 0$ . (If  $P_{\max} = 0$  then  $P_0 = 0$  and  $X_0 = X^*$ , so the trajectory is already at equilibrium and there is nothing to prove; take  $B = 1$ .) No lower bound on  $\mu$  is required. There are exactly two constraints on  $\epsilon$ : the bound  $\epsilon \leq 1/4$  used for (23)–(24), and positivity of  $\Gamma(\epsilon)$ . We may therefore fix

$$\epsilon := \min\left\{\frac{1}{4}, \frac{\gamma - K_1}{2\Lambda}\right\}, \quad (31)$$

which gives  $\Gamma(\epsilon) \geq \frac{\gamma - K_1}{2} \geq \frac{\gamma}{4} > 0$ .

Shrinking  $B$  does not affect the rate, since by the preceding paragraph the auxiliary decay rate  $\alpha$  is  $B$ -invariant; it enters only the prefactor. Inverting (23) for each quantity separately gives

$$f(X) - f(X^*) \leq \mathcal{W}_{\epsilon}, \quad \|P\|_F^2 \leq 4\mathcal{W}_{\epsilon}, \quad \|R\|^2 + \|C\|^2 \leq \frac{2}{\mu B}\mathcal{W}_{\epsilon}, \quad \|X - X^*\|_F^2 \leq \frac{4}{\epsilon}\mathcal{W}_{\epsilon},$$

and since the conclusion of Theorem 2 is the *sum* of these, the four bounds add:

$$C_{\star} := \left(5 + \frac{2}{\mu B} + \frac{4}{\epsilon}\right)\mathcal{W}_{\epsilon}(0). \quad (32)$$

Consequently, there exists  $\kappa_0 > 0$  such that

$$\dot{\mathcal{W}}_{\epsilon} \leq -\kappa_0\left(f(X) - f(X^*) + \|P\|_F^2 + \|R\|^2 + \|C\|^2 + \|X - X^*\|_F^2\right).$$

Using the upper bound (24), we obtain  $\dot{\mathcal{W}}_{\epsilon} \leq -\kappa\mathcal{W}_{\epsilon}$  for some  $\kappa > 0$ . Grönwall's inequality then gives  $\mathcal{W}_{\epsilon}(t) \leq \mathcal{W}_{\epsilon}(0)e^{-\kappa t}$ , and the lower bound (23) implies exponential decay of each individual term.  $\square$

### E.3 Convergence and algebraic rates at $\gamma = 0$

We now treat the undamped case  $\gamma = 0$ , which is the variant recommended in practice and used in all experiments. Two results are established, of quite different character. The first is that the dynamics converge at all: for any coercive  $f$  with a unique critical point, every trajectory tends to the minimizer, with no restriction on  $\alpha$ ,  $\mu$  or  $\epsilon_{\text{stab}}$  (Theorem 3). The second is a rate: under uniform strong convexity,

$$\mathcal{E}(t) = \mathcal{O}(t^{-1}) \quad (\epsilon_{\text{stab}} = 0), \quad \mathcal{E}(t) = \mathcal{O}(t^{-1/2}) \quad (\epsilon_{\text{stab}} > 0),$$

with all constants explicit in  $m, n, m_f, M_f, \alpha, \mu, \epsilon_{\text{stab}}$  and  $\mathcal{E}(0)$  (Theorem 8). The decay is algebraic, not exponential, and the two exponents correspond to two regimes separated by the size of the smoothed friction relative to  $\epsilon_{\text{stab}}$ .

**Why the rate is delicate.** At  $\gamma = 0$  the only damping is  $\tilde{\xi} \odot P$ , and  $\tilde{\xi}$  is built from *past* momentum. Two consequences follow. First, as  $P \rightarrow 0$  the friction switches itself off, so no exponential rate can hold and the answer must be algebraic. Second,  $\dot{\mathcal{E}} = -\langle P, \tilde{\xi} \odot P \rangle_F$  vanishes at every turning point of the oscillation, where  $P = 0$ ; there is therefore no differential inequality for  $\mathcal{E}$  at a single instant, and every estimate must be made over a stretch of time. That is why the argument is organized into blocks, with a three-way alternative on each: either the energy drops by a fixed factor, or a scalar moving average contracts toward the energy scale, or a finite-window concentration inequality forces a definite increase of a reciprocal-energy Lyapunov function.

**Notation for this subsection.** Throughout,  $\gamma = 0$  and  $\epsilon_{\text{stab}} \geq 0$  is written  $\epsilon_{\text{stab}}$ ; we abbreviate  $S = \mathbf{1}_m^\top R$  and use the reduced energy  $\mathcal{E}$  of Lemma 1, writing  $\mathcal{E}_0 := \mathcal{E}(0)$ , so that  $P_{\text{max}}^2 = 2\mathcal{E}_0$ . Set

$$A := [P]^2, \quad \sigma := \sum_{ij} A_{ij} = \|P\|_F^2, \quad (33)$$

and let  $W$  solve the entrywise exponential moving average

$$W' = \alpha(A - W), \quad W(0) = 0. \quad (34)$$

Write

$$\bar{a}_i = \sum_j W_{ij}, \quad \bar{b}_j = \sum_i W_{ij}, \quad z = \sum_{ij} W_{ij} = \alpha\mu S, \quad \vartheta := \alpha\mu \epsilon_{\text{stab}}. \quad (35)$$

Solving (17c)–(17d) by variation of constants identifies these with the friction factors,

$$R_i = \frac{\bar{a}_i}{\alpha\mu}, \quad C_j = \frac{\bar{b}_j}{\alpha\mu}, \quad S = \frac{z}{\alpha\mu}, \quad (36)$$

so  $z$  is the total smoothed friction. When  $\epsilon_{\text{stab}} = 0$  we set  $\tilde{\xi} = 0$  at  $S = 0$ ; with  $R(0) = C(0) = 0$  the identity  $\mathbf{1}_m^\top R = \mathbf{1}_n^\top C$  is invariant, so this is the continuous extension along the trajectory. To avoid collision with the symbols of Theorem 2 we write  $\beta$  for the upper band constant,  $\nu := mn$ , and  $K_*$  for the concentration constant.

#### E.3.1 Unconditional convergence

**Theorem 3** (Convergence at  $\gamma = 0$ ). *Suppose  $f \in C^2$  is coercive and has a unique critical point  $X^*$ . Let  $\alpha, \mu > 0$  and  $\epsilon_{\text{stab}} > 0$ . Then every solution of (17a)–(17d) with  $\gamma = 0$  and  $R(0), C(0) \geq 0$  satisfies*

$$(X(t), P(t), R(t), C(t)) \longrightarrow (X^*, 0, 0, 0).$$

*The same conclusion holds for  $\epsilon_{\text{stab}} = 0$  along the trace-matched solution whenever the continuous extension above is used.*

*Proof.* Lemma 1 makes the trajectory precompact, and (20) with  $\gamma = 0$  gives  $\dot{\mathcal{E}} = -D \leq 0$  where  $D := \langle P, \tilde{\xi} \odot P \rangle_F$ , so LaSalle's invariance principle applies. Let  $(X, P, R, C)$  be a bounded complete trajectory in the largest invariant subset of  $\{D = 0\}$ . Bounded completeness removes the backward-growing homogeneous mode from (17c)–(17d), leaving

$$R_i(t) = \frac{1}{\mu} \int_{-\infty}^t e^{-\alpha(t-s)} \sum_j P_{ij}(s)^2 ds, \quad C_j(t) = \frac{1}{\mu} \int_{-\infty}^t e^{-\alpha(t-s)} \sum_i P_{ij}(s)^2 ds.$$

If  $P_{ij}(t_0) \neq 0$  for some  $i, j, t_0$ , continuity makes both integrands positive on an interval just before  $t_0$ , hence  $R_i(t_0)C_j(t_0) > 0$  and therefore  $\tilde{\xi}_{ij}(t_0)P_{ij}(t_0)^2 > 0$ , contradicting  $D = 0$ . Thus  $P \equiv 0$ , and (17b) then gives  $\nabla f(X) = 0$ , so  $X = X^*$  by uniqueness of the critical point. Finally the only bounded complete solutions of  $\dot{R} = -\alpha R$  and  $\dot{C} = -\alpha C$  are identically zero.  $\square$

### E.3.2 A quantitative observability window

For the rate we assume uniform strong convexity and smoothness,

$$0 < m_f I \preceq \nabla^2 f(X) \preceq M_f I \quad \text{for every } X, \quad (37)$$

which implies (21) with  $a = m_f/M_f$  and  $b = m_f/2$ . Write  $Y = X - X^*$  and set

$$K_f = 1 + \frac{1}{m_f}, \quad C_1 = \frac{1}{a} + \frac{1}{2}, \quad C_2 = \frac{2K_f}{a}, \quad C_3 = \frac{\xi_{\max}}{2ab}, \quad T_0 = 4(C_2 + C_3), \quad c_0 = \frac{1}{2C_1}. \quad (38)$$

**Lemma 4** (Window observability). *For every  $t \geq 0$ , either*

$$\mathcal{E}(t + T_0) \leq \frac{1}{2} \mathcal{E}(t), \quad (39)$$

or

$$\int_t^{t+T_0} \|P(s)\|_F^2 ds \geq c_0 T_0 \mathcal{E}(t + T_0). \quad (40)$$

*Proof.* For  $V = \langle Y, P \rangle_F$  we have  $\dot{V} = \|P\|_F^2 - \langle Y, \nabla f(X) \rangle_F - \langle Y, \tilde{\xi} \odot P \rangle_F$ . Integrate over  $[t, t + T_0]$ . By Cauchy–Schwarz against the measure  $\tilde{\xi}$ , then in time, then Young, using (19) and  $\int D = \mathcal{E}(t) - \mathcal{E}(t + T_0)$ ,

$$\int |\langle Y, \tilde{\xi} \odot P \rangle_F| \leq \left( \int D \right)^{1/2} \left( \xi_{\max} \int \|Y\|_F^2 \right)^{1/2} \leq \frac{b}{2} \int \|Y\|_F^2 + \frac{\xi_{\max}}{2b} (\mathcal{E}(t) - \mathcal{E}(t + T_0)).$$

Also  $|V| \leq K_f \mathcal{E}$ , since  $\|Y\|_F^2 \leq 2f/m_f$  and  $f \leq \mathcal{E}$ . Using (21), absorbing the  $\|Y\|_F^2$  terms and substituting  $\int f = \int \mathcal{E} - \frac{1}{2} \int \|P\|_F^2$  gives

$$\int_t^{t+T_0} \mathcal{E} \leq C_1 \int_t^{t+T_0} \|P\|_F^2 + C_2 \mathcal{E}(t) + C_3 (\mathcal{E}(t) - \mathcal{E}(t + T_0)).$$

If (39) fails then  $\mathcal{E}(t) < 2\mathcal{E}(t + T_0)$ , so the last two terms are at most  $2(C_2 + C_3)\mathcal{E}(t + T_0)$ . Since  $\mathcal{E}$  is nonincreasing the left side is at least  $T_0 \mathcal{E}(t + T_0)$ , and the choices (38) give (40).  $\square$

### E.3.3 Finite-window concentration

**Lemma 5** (Tracked block concentration). *Let  $I = [u, u + L_1]$  and suppose that for some  $e, \ell, \beta > 0$ ,*

$$\ell e \leq z(t) \leq \beta e \quad (t \in I). \quad (41)$$

If

$$L_1 \geq \max \left\{ \frac{1}{2\alpha}, \frac{\nu\beta^2}{\alpha\ell^2} \right\}, \quad (42)$$

then

$$\int_I \sum_{ij} \frac{A_{ij} \bar{a}_i \bar{b}_j}{z} dt \geq \frac{\ell^4}{8\nu^2\beta^2} L_1 e^2. \quad (43)$$

*Proof.* Set  $F = \sum_{ij} W_{ij}^2$  and  $H_W = \sum_{ij} A_{ij} W_{ij}$ . Since  $\bar{a}_i, \bar{b}_j \geq W_{ij}$ , two applications of Cauchy–Schwarz give

$$\int_I \sum_{ij} \frac{A_{ij} \bar{a}_i \bar{b}_j}{z} \geq \int_I \frac{H_W^2}{\sigma z} \geq \frac{(\int_I H_W)^2}{\int_I \sigma z}.$$

From (34),  $F' = 2\alpha(H_W - F)$  and  $(z^2)' = 2\alpha(\sigma z - z^2)$ , giving the exact finite-window identities

$$\int_I H_W = \int_I F + \frac{F(u + L_1) - F(u)}{2\alpha}, \quad \int_I \sigma z = \int_I z^2 + \frac{z(u + L_1)^2 - z(u)^2}{2\alpha}.$$

Because  $F \geq z^2/\nu$  and  $F(u) \leq z(u)^2$ , the hypotheses (41)–(42) imply  $\int_I H_W \geq L_1 \ell^2 e^2 / (2\nu)$  and  $\int_I \sigma z \leq 2L_1 \beta^2 e^2$ ; the first uses  $L_1 \geq \nu\beta^2 / (\alpha\ell^2)$  and the second  $L_1 \geq 1/(2\alpha)$ . Substitution proves (43).  $\square$

### E.3.4 Tracking over blocks

Set

$$\ell = \alpha c_0 T_0 e^{-\alpha T_0}, \quad \beta = 36, \quad \nu = mn, \quad L_1 = \max\left\{\frac{1}{2\alpha}, \frac{\nu\beta^2}{\alpha\ell^2}, \frac{\log 4}{\alpha}\right\}, \quad L = T_0 + L_1. \quad (44)$$

Let  $t_k = kL$ ,  $\mathcal{E}_k = \mathcal{E}(t_k)$  and  $z_k = z(t_k)$ . If  $\mathcal{E}_K = 0$  for some  $K$  then  $\mathcal{E}$ , being nonnegative and nonincreasing, vanishes on  $[t_K, \infty)$  and every bound below is trivial there; we therefore assume  $\mathcal{E}_k > 0$  for all  $k$  and set  $r_k = z_k/\mathcal{E}_k$ . A block is *halving* when  $\mathcal{E}_{k+1} \leq \mathcal{E}_k/2$ , *untracked* when it is not halving but  $r_k > 16$ , and *controlled* otherwise.

**Lemma 6** (Tracking alternative). *On every controlled block, with  $e = \mathcal{E}_{k+1}$ ,*

$$\ell e \leq z(t) \leq \beta e \quad (t \in [t_k + T_0, t_{k+1}]), \quad (45)$$

so Lemma 5 applies on the trimmed interval. Moreover, among the first  $K$  blocks the number  $U_K$  of untracked blocks satisfies

$$U_K \leq C_{\text{tr}} \left(1 + \log \frac{\mathcal{E}_0}{\mathcal{E}_K}\right), \quad C_{\text{tr}} = \frac{1}{\log \frac{8}{7}} \left(1 + \frac{2 \log \frac{3}{2}}{\log 2} + \log \frac{3}{2}\right). \quad (46)$$

*Proof.* Suppose block  $k$  is not halving, so  $\mathcal{E}_{k+1} > \mathcal{E}_k/2$ . Every window  $[t - T_0, t] \subset [t_k, t_{k+1}]$  also fails to halve, because  $\mathcal{E}(t) \geq \mathcal{E}_{k+1} > \mathcal{E}_k/2 \geq \mathcal{E}(t - T_0)/2$ . Hence Lemma 4 and  $z' = \alpha(\|P\|_F^2 - z)$  give, for  $t \geq t_k + T_0$ ,

$$z(t) \geq \alpha e^{-\alpha T_0} \int_{t-T_0}^t \|P(s)\|_F^2 ds \geq \ell \mathcal{E}(t) \geq \ell \mathcal{E}_{k+1}.$$

If the block is controlled then  $z_k \leq 16\mathcal{E}_k$ , while  $\|P\|_F^2 \leq 2\mathcal{E}_k$  throughout, so  $z(t) \leq z_k + 2\mathcal{E}_k \leq 18\mathcal{E}_k < 36\mathcal{E}_{k+1}$ , proving (45). Note that  $\ell$  depends on the observability window  $T_0$  and not on the block length  $L_1$ : had it been obtained from decay across the whole block it would behave like  $e^{-\alpha L_1}$ , while (42) simultaneously demands  $L_1 \gtrsim 1/\ell^2$ , and the two requirements could not both be met. Separating the two lengths removes the conflict.

For the count, put  $q = e^{-\alpha L} \leq 1/4$ , which holds because  $L \geq L_1 \geq (\log 4)/\alpha$ . Since  $\|P\|_F^2 \leq 2\mathcal{E}_k$  on the block, variation of constants gives

$$z_{k+1} \leq q z_k + 2(1 - q)\mathcal{E}_k, \quad \text{hence} \quad r_{k+1} \leq d_k \left(\frac{r_k}{4} + 2\right), \quad d_k = \frac{\mathcal{E}_k}{\mathcal{E}_{k+1}} \geq 1. \quad (47)$$

(a) *Non-halving blocks.* Here  $d_k < 2$ , so  $r_{k+1} \leq \frac{1}{2}r_k + 4$ . If  $r_k > 16$  then  $r_{k+1} \leq \frac{3}{4}r_k$ ; if  $r_k \leq 16$  then  $r_{k+1} \leq 12$ . Thus a controlled block is followed, unless the next block halves, by another controlled block, so a maximal run of untracked blocks can be preceded only by a halving block. Since  $r_0 = 0$ , no untracked run begins at  $k = 0$ , and the number of untracked runs is at most  $|\mathcal{H}|$ , where  $\mathcal{H}$  is the set of halving blocks.

(b) *Halving blocks.* From (47),  $1 + \frac{r_{k+1}}{16} \leq 1 + \frac{d_k(r_k+8)}{64} \leq d_k \left(1 + \frac{r_k}{16}\right)$ , the second inequality reducing to  $64 \leq 56d_k + 3d_k r_k$ , which holds since  $d_k \geq 2 \geq 8/7$ .

(c) *Controlled blocks.* By (a),  $r_{k+1} \leq \max\{r_k, 8\}$ .

Now set  $\psi_k = \log(1 + r_k/16) \geq 0$ , so  $\psi_0 = 0$ . On an untracked block (a) gives  $\psi_{k+1} - \psi_k \leq \log \frac{3/4 \cdot 16 + 16}{16 + 16} = -\log \frac{8}{7}$ , since  $r \mapsto \log \frac{1+3r/64}{1+r/16}$  is nonincreasing and is evaluated at worst at  $r_k = 16$ . On a halving block (b) gives  $\psi_{k+1} - \psi_k \leq \log d_k$ . On a controlled block (c) gives  $\psi_{k+1} \leq \max\{\psi_k, \log \frac{3}{2}\}$ , so  $\psi$  can increase only up to the level  $\log \frac{3}{2}$  and the net increase over any maximal run of controlled blocks is at most  $\log \frac{3}{2}$ , however long the run; this is the point at which summing the increases block by block would fail, since  $\psi$  does increase on individual controlled blocks. The number of maximal controlled runs is at most  $1 + |\mathcal{H}| + (\text{untracked runs}) \leq 1 + 2|\mathcal{H}|$ . Summing over the first  $K$  blocks and using  $\psi_K \geq 0 = \psi_0$ ,

$$U_K \log \frac{8}{7} \leq \sum_{k \in \mathcal{H}} \log d_k + (1 + 2|\mathcal{H}|) \log \frac{3}{2}.$$

Finally  $\prod_{k < K} d_k = \mathcal{E}_0/\mathcal{E}_K$  with every  $d_k \geq 1$ , so  $\sum_{k \in \mathcal{H}} \log d_k \leq \log(\mathcal{E}_0/\mathcal{E}_K)$  and  $|\mathcal{H}| \leq \log(\mathcal{E}_0/\mathcal{E}_K)/\log 2$ , which gives (46).  $\square$



### E.3.5 The algebraic rate

Define

$$K_\star = \frac{\ell^5 L_1}{8\nu^2 \beta^2 \alpha \mu}. \quad (48)$$

**Lemma 7** (Reciprocal-energy increment). *Let  $\Phi(e) = \frac{\beta}{e} + \frac{\vartheta}{2e^2}$ . On every controlled block,*

$$\Phi(\mathcal{E}_{k+1}) - \Phi(\mathcal{E}_k) \geq \frac{K_\star}{8}. \quad (49)$$

*Proof.* On  $I_k = [t_k + T_0, t_{k+1}]$ , which has length  $L_1$ , Lemma 6 supplies the band (41) with  $e = \mathcal{E}_{k+1}$ , so Lemma 5 applies. From (36),

$$D = \frac{1}{\alpha \mu} \sum_{ij} \frac{A_{ij} \bar{a}_i \bar{b}_j}{z + \vartheta}.$$

Since  $z/(z + \vartheta) \geq \ell \mathcal{E}_{k+1}/(\beta \mathcal{E}_{k+1} + \vartheta)$  on  $I_k$ , combining with (43) gives

$$\mathcal{E}_k - \mathcal{E}_{k+1} \geq K_\star \frac{\mathcal{E}_{k+1}^3}{\beta \mathcal{E}_{k+1} + \vartheta}. \quad (50)$$

The block is non-halving, so  $\mathcal{E}_k < 2\mathcal{E}_{k+1}$ , and for  $s \in [\mathcal{E}_{k+1}, \mathcal{E}_k]$ ,

$$-\Phi'(s) = \frac{\beta}{s^2} + \frac{\vartheta}{s^3} \geq \frac{\beta \mathcal{E}_{k+1} + \vartheta}{8\mathcal{E}_{k+1}^3}.$$

Integrating this over  $[\mathcal{E}_{k+1}, \mathcal{E}_k]$  and using (50) proves (49).  $\square$

**Theorem 8** (Parameter-dependent algebraic rate). *Assume (37),  $\gamma = 0$  and  $R(0) = C(0) = 0$ , and let the constants be as in (38)–(48). There are numerical constants  $c, C > 0$  such that*

$$\mathcal{E}(t) \leq C \left[ \mathcal{E}_0 e^{-ct/L} + \frac{\beta L}{K_\star(L+t)} + \sqrt{\frac{\vartheta L}{K_\star(L+t)}} \right] \quad (t \geq 0). \quad (51)$$

*Consequently  $\mathcal{E}(t) = \mathcal{O}(t^{-1})$  when  $\epsilon_{\text{stab}} = 0$  and  $\mathcal{E}(t) = \mathcal{O}(t^{-1/2})$  when  $\epsilon_{\text{stab}} > 0$ . All constants depend only on  $m, n, m_f, M_f, \alpha, \mu, \epsilon_{\text{stab}}$  and  $\mathcal{E}_0$  through the displayed formulas.*

*Proof.* Consider the first  $K$  blocks and let  $H_K, U_K, M_K$  count the halving, untracked and controlled ones. Every halving block contributes at least  $\log 2$  to  $\log(\mathcal{E}_0/\mathcal{E}_K)$ , and Lemma 6 bounds  $U_K$ . If  $H_K + U_K \geq K/2$  these two facts force  $\log(\mathcal{E}_0/\mathcal{E}_K) \gtrsim K$ , hence  $\mathcal{E}_K \leq C\mathcal{E}_0 e^{-cK}$  for numerical  $c, C > 0$ .

Otherwise  $M_K \geq K/2$ . Since  $\mathcal{E}$  is nonincreasing and  $\Phi$  is decreasing,  $\Phi(\mathcal{E}_k)$  is nondecreasing in  $k$ , so summing Lemma 7 over the controlled blocks yields  $\Phi(\mathcal{E}_K) \geq K_\star K/16$ . At least one of the two terms of  $\Phi$  is at least half of this, whence

$$\mathcal{E}_K \leq \max \left\{ \frac{32\beta}{K_\star K}, 4\sqrt{\frac{\vartheta}{K_\star K}} \right\}.$$

To pass to continuous time fix  $t \geq 0$  and put  $K = \lfloor t/L \rfloor$ . If  $K = 0$  then  $t < L$ , so  $e^{-ct/L} \geq e^{-c}$  and  $\mathcal{E}(t) \leq \mathcal{E}_0 \leq e^c \mathcal{E}_0 e^{-ct/L}$ , which is absorbed by enlarging  $C$ . If  $K \geq 1$  then  $t_K = KL \leq t$ , so  $\mathcal{E}(t) \leq \mathcal{E}_K$ ; moreover  $\lfloor x \rfloor \geq x/2$  for  $x \geq 1$  gives  $K \geq t/(2L)$ , and  $t \geq L$  gives  $L+t \leq 2t$ , so  $1/K \leq 2L/t \leq 4L/(L+t)$ . Substituting turns the three alternatives into the three displayed terms of (51) after renaming numerical constants.  $\square$

**Remark 4** (Scope). *The estimates are upper bounds and do not preclude faster decay along particular trajectories. Formal averaging predicts the sharper equivalents*

$$\mathcal{E}(t) \sim \frac{\alpha \mu}{t} \quad (\epsilon_{\text{stab}} = 0), \quad \mathcal{E}(t) \sim \alpha \mu \sqrt{\frac{\epsilon_{\text{stab}}}{2t}} \quad (\epsilon_{\text{stab}} > 0),$$

*with the crossover between the two regimes at  $\mathcal{E} \approx \alpha \mu \epsilon_{\text{stab}}$ ; proving these equivalents is a separate averaging problem. Appendix E.3.6 confirms both exponents, the crossover scale and the predicted dependence on  $\alpha, \mu$  and  $\epsilon_{\text{stab}}$  numerically.*

### E.3.6 Numerical validation

The exponents of Theorem 8 and the parameter *relations* predicted by Remark 4 were both checked numerically on quadratic objectives  $f(X) = \frac{1}{2}\langle X, \mathcal{A}X \rangle_F$  with  $\mathcal{A}$  a random symmetric positive definite operator on  $\mathbb{R}^{mn}$ , started from  $R(0) = C(0) = 0$ .

Two independent integrators were used. Parameter sweeps were run with an adaptive Runge–Kutta method (ode89, relative tolerance  $10^{-10}$ , absolute tolerance  $10^{-12}$ ). The long-horizon runs used the BACD operator splitting of the method itself (Appendix A) at fixed step  $h$ , which is the natural integrator here: each sub-flow is solved exactly, non-negativity of  $R$  and  $C$  is preserved by construction, no error control is required, and horizons of  $10^7$  time units are inexpensive. On their common range the two agree to about 0.01 in every measured slope, and halving  $h$  from 0.02 to 0.01 changes every slope by less than  $10^{-3}$ .

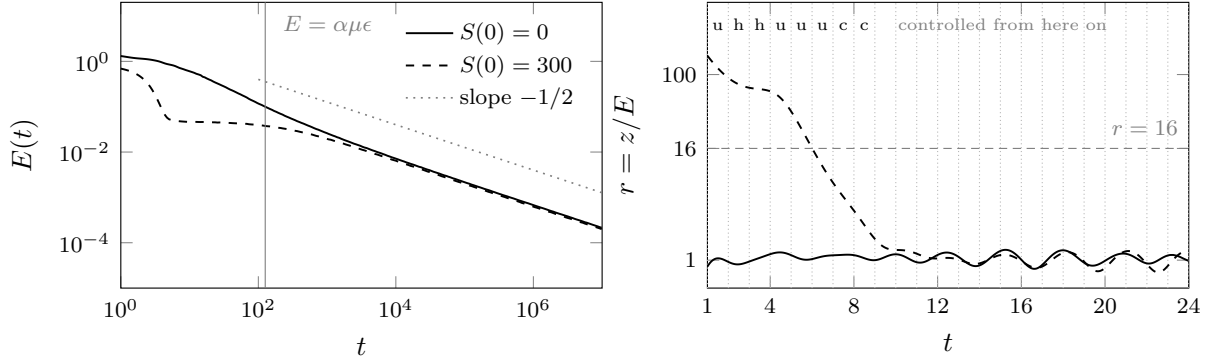


Figure 4: Decay and the block classification of Lemma 6, for  $m = 4, n = 3, \epsilon_{\text{stab}} = 0.1, \alpha = \mu = 1$ , integrated with the BACD splitting to  $t = 10^7$ . *Left*: two trajectories, the standard initialization  $R(0) = C(0) = 0$  (solid) and one started with a large initial friction,  $\mathbf{1}_m^\top R = \mathbf{1}_n^\top C = 300$  (dashed). Both settle onto the predicted  $t^{-1/2}$  decay, with the crossover from the first regime occurring near  $\mathcal{E} = \alpha\mu\epsilon_{\text{stab}}$ . *Right*: a linear-time detail of the ratio  $r = z/\mathcal{E}$  that decides the classification, with dotted vertical lines marking the block boundaries and the letters above giving the class of each block of the second trajectory: u untracked, h halving, c controlled. The standard trajectory has  $r \approx 1$  from the outset, so all of its blocks are controlled. The second begins at  $r \approx 160$ , spends its first blocks untracked, passes through two halving blocks, and is controlled from the eighth onwards.

**Exponents are as predicted, and independent of dimension.** Local slopes of  $\log \mathcal{E}$  against  $\log t$  on successive decades, integrated to  $T = 10^7$  at  $h = 0.02$  with  $\alpha = \mu = 1$ :

|                                                    | $[10^3, 10^4]$ | $[10^4, 10^5]$ | $[10^5, 10^6]$ | $[10^6, 10^7]$ |
|----------------------------------------------------|----------------|----------------|----------------|----------------|
| $\epsilon_{\text{stab}} = 0$ (predicted $-1$ )     |                |                |                |                |
| $m = n = 2$                                        | -0.955         | -0.980         | -0.986         | -0.989         |
| $m = 4, n = 3$                                     | -0.991         | -0.996         | -0.996         | -0.998         |
| $m = 6, n = 5$                                     | -0.993         | -0.997         | -0.998         | -0.999         |
| $\epsilon_{\text{stab}} = 0.1$ (predicted $-1/2$ ) |                |                |                |                |
| $m = n = 2$                                        | -0.499         | -0.497         | -0.496         | -0.497         |
| $m = 4, n = 3$                                     | -0.558         | -0.518         | -0.505         | -0.501         |
| $m = 6, n = 5$                                     | -0.593         | -0.530         | -0.509         | -0.503         |

Both exponents are independent of the matrix dimensions, as Theorem 8 requires. The approach to  $-1/2$  is slower for larger  $mn$ , and the crossover relation explains why: the second regime begins only once  $\mathcal{E} \lesssim \alpha\mu\epsilon_{\text{stab}}$ , and the constant in the first-regime law  $\mathcal{E} \approx c_1\alpha\mu/t$  grows with  $mn$ , so the crossing occurs later for larger systems. On a horizon of  $2 \cdot 10^4$  the  $6 \times 5$  system still reads  $-0.583$ ; by  $[10^6, 10^7]$  it has reached  $-0.503$ .

**The predicted dependence on  $\alpha, \mu$  and  $\epsilon_{\text{stab}}$  holds.** If the asymptotics are  $\mathcal{E} \sim c_1\alpha\mu/t$  and  $\mathcal{E} \sim c_2\alpha\mu\sqrt{\epsilon_{\text{stab}}/2t}$ , the ratios below are constant. Evaluated at  $t = 10^4$  with  $m = 3, n = 2$ :

| quantity                                                                                     | parameters varied                                                              | mean | spread (max/min) |
|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|------|------------------|
| $t\mathcal{E}/(\alpha\mu), \epsilon_{\text{stab}} = 0$                                       | $\alpha, \mu \in \{\frac{1}{2}, 1, 2\}$                                        | 4.69 | 1.29             |
| $\mathcal{E}\sqrt{t}/(\alpha\mu\sqrt{\epsilon_{\text{stab}}/2}), \epsilon_{\text{stab}} > 0$ | $\alpha, \mu \in \{\frac{1}{2}, 1\}, \epsilon_{\text{stab}} \in \{0.05, 0.2\}$ | 2.31 | 1.14             |
| $\mathcal{E}/(\alpha\mu\epsilon_{\text{stab}})$ at the crossing                              | $\alpha, \mu \in \{\frac{1}{2}, 1\}, \epsilon_{\text{stab}} \in \{0.05, 0.2\}$ | 0.87 | 1.21             |

In the first row  $\alpha$  and  $\mu$  each range over a factor of 4, so  $\alpha\mu$  ranges over a factor of 16, yet the ratio moves by less than 30%. The third row locates the transition between the two regimes at  $\mathcal{E} \approx 0.87 \alpha\mu \epsilon_{\text{stab}}$ , confirming the crossover scale. For the scalar system  $m = n = 1$  with  $\alpha = \mu = 1$ , evaluating at  $t = 10^5$  gives  $t\mathcal{E}/(\alpha\mu) = 0.909$  and  $\mathcal{E}\sqrt{t}/(\alpha\mu\sqrt{\epsilon_{\text{stab}}/2}) = 0.884$ , both within about 10% of the value 1 predicted by formal averaging; for  $m, n > 1$  the same relations hold with a dimension-dependent constant, which the averaging computation does not predict.

Since  $\epsilon_{\text{stab}}$  is small in practice, the crossover at  $\mathcal{E} \approx \alpha\mu \epsilon_{\text{stab}}$  lies far below any energy reached in a finite run, so the  $t^{-1}$  law of the first regime is the operative one at any horizon of practical interest.

## F Related Work: Algebraic Rates and Factored Optimizers

**Damping determines the rate.** For  $\ddot{x} + \gamma\dot{x} + \nabla f(x) = 0$  with constant  $\gamma > 0$  and strongly convex  $f$ , function values converge exponentially; this is the regime of Theorem 2. The assumption is not cosmetic: at  $\gamma = 0$  the system is conservative, the energy is exactly preserved, and no trajectory converges. The standard route to algebraic rates therefore keeps the damping but lets it vanish on a schedule. Su et al. [19] identified  $\ddot{x} + \frac{3}{t}\dot{x} + \nabla f(x) = 0$  as the continuous limit of Nesterov’s method and read its  $\mathcal{O}(t^{-2})$  rate off the ODE; Attouch et al. [2] analyzed  $\ddot{x} + \frac{\alpha}{t}\dot{x} + \nabla f(x) = 0$  in general, with  $\alpha \geq 3$  giving  $f - \min f \leq C/t^2$ . In both the damping is an explicit function of time chosen by the analyst. Our damping is neither constant nor scheduled:  $\xi \odot P$  is generated by the trajectory from past momentum, vanishing as  $P \rightarrow 0$ , which precludes an exponential rate, but never identically zero, which is why convergence survives the removal of  $\gamma$ . Being a function of state rather than time, it falls outside both arguments, and the block construction of Appendix E.3 replaces them.

**Factored optimizers.** Adafactor [17] introduced rank-1 factorization of optimizer state; Hong and Lin [4] give a discrete-time convergence theory for it in the non-convex smooth setting, treating it as an approximation to Adam and obtaining rates for stationarity. Closest to the present analysis, Nguyen et al. [14] study rank-1 parameterized momentum and scaling estimators in continuous time, writing a factored Adafactor variant as an ODE and deducing convergence to critical points from a monotonically decreasing Hamiltonian. Two differences delimit our contribution: their analysis gives descent and asymptotic convergence but no rate, and it requires strictly positive damping, whereas Theorem 8 gives an algebraic rate with explicit constants at  $\gamma = 0$ , the regime the method is deployed in. To our knowledge it is the first convergence rate for a rank-1 factored optimizer in continuous time, and the first such analysis without positive damping. The block construction does require momentum, so it does not apply to Adafactor as originally proposed, whose continuous limit is first order in  $X$ ; it may however be useful for Adafactor-m.

**Friction-based methods.** R-iKFAD descends from global scalar friction [6] and per-parameter friction in iKFAD [7]. As noted in Section 3, at  $m = n = 1$  with  $\epsilon_{\text{stab}} = 0$  the dynamics reduce exactly to the scalar case, so Theorem 8 yields a  $t^{-1}$  rate there too. For full-friction iKFAD at  $\gamma = 0$  the block architecture applies with the concentration step removed, since elementwise friction makes the dissipation directly available; that version carries no normalizing denominator and hence no  $\epsilon_{\text{stab}}$ , so we expect only the  $t^{-1}$  law. We have not carried out that analysis here.